INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Infosys SE Complete Preparation — Time & Work, Efficiency, Pipes & Cisterns

Section: prepinsta infosys SE | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Infosys SE Complete Preparation — Time & Work, Efficiency, Pipes & Cisterns

Source Portal: PrepInsta Infosys Preparation Suite (`https://prepinsta.com/infosys/aptitude/time-work/`)
Target Role: Infosys Systems Engineer (SE) | **Category:** 01_Mathematical_Ability_Technical_Aptitude
Total Verified Questions Extracted: 10
Quizzes Covered: quiz-1, quiz-2, quiz-3, quiz-4, quiz-5, quiz-6

📋 Section 1: Rules to Keep in Mind

These rules and fundamental principles govern all Infosys questions on this topic:

- If A can complete a work in n days, A's 1 day work = 1/n.
- Total Work = Efficiency * Time.
- Combined work of A and B: 1/Time = 1/A + 1/B => Time = (A * B) / (A + B).
- Pipes and Cisterns: Inlet pipe rate is positive (+1/x); Outlet / leak pipe rate is negative (-1/y).
- Net rate = 1/x - 1/y.

⚡ Section 2: How to Solve Quickly

Proven time-saving strategies recommended by PrepInsta for Infosys 2026 examination:

1. **Option Elimination via Unit Digits & Multiples:** Before doing full calculations, inspect the options to see if last digits or divisibility by 3, 7, 9, 11 can eliminate 2 to 3 wrong choices.
2. **Approximation & Magnitude Checks:** When exact arithmetic is tedious, round numbers to neat multiples (e.g., 99 ≈ 100) to find the nearest valid option.
3. **Reverse Option Substitution:** In complex algebraic equations, plug option values directly into the question stem to test validity.

🚀 Section 3: Tricks & Shortcuts

Key mathematical formulas, mental-math shortcuts, and speed axioms:

- LCM Method for Total Work: Assume total work = LCM of individual times; find daily unit efficiency.
- Chain Rule: (M1 * D1 * H1) / W1 = (M2 * D2 * H2) / W2.

- Efficiency is inversely proportional to time taken: Ratio of efficiency A:B = y:x if time ratio is x:y.

📚 Section 4: Practice Questions, Options & Step-by-Step Solutions

Contains **10** verbatim questions with options, verified answer keys, and complete derivations:

Question 1 (PrepInsta quiz-1 - Q1)

Question Stem:
If Rita spends every day 40 minutes for watering the plants how much time does Rita spend watering the plants in 20 days?

Options:
- **(A)** `12hours`
- **(B)** `13.33hours`
- **(C)** `12.5hours`
- **(D)** `15.5hours`
- **(E)** `14.33hours`

Correct Answer: (B) `13.33hours`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
sita 1 day sepend time =40 min


sita 20 days spend time =20*40min


=800 min


so, convert it into hours= 800/60 hrs


13.33hrs
```

Question 2 (PrepInsta quiz-1 - Q2)

Question Stem:
If a pipe A can fill a tank in 40 minutes and pipe B fill the same tank in 30 minutes. How long will it take for both pipes together to fill the tank?

Options:
- **(A)** `17`
- **(B)** `16`
- **(C)** `15`
- **(D)** `120/7`
- **(E)** `13`

Correct Answer: (D) `120/7`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Time taken by both pipes together to fill the tank = 1/(1/40 +1/30) = 120/7 min
```

Question 3 (PrepInsta quiz-1 - Q3)

Question Stem:

3 men finish painting a wall in 8 days. Four boys do the same job in 7 days. In how many days will 2 men and 2 boys working together paint two such walls of the same size?

Options:

- **(A)** `6 6/13 days`
- **(B)** `3 3/5 days`
- **(C)** `9 2/5 days`
- **(D)** `12 12/13 days`

Correct Answer: (D) `12 12/13 days`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
3men 8days --&gt;1 man=1/24 days(1/3*1/8)

4boys 7days--&gt;1 boy=1/28 days

2men+2boys==2/24+2/28--&gt;1/12+1/14

2 such walls--&gt;1/24+1/28

take LCM, 13/168

reciprocal of it, 168/13

i.e,12 12/13
```

Question 4 (PrepInsta quiz-1 - Q4)

Question Stem:

If A and B can do a piece of work in 7.5 days. If B works 1/2 of work and remaining work was completed by A, taking total time of 20 days to complete the work. If b is more efficient then B can do work in how many days?

Options:

- **(A)** `11`
- **(B)** `10`

- **(C)** `20`
- **(D)** `8`

Correct Answer: (B) `10`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
10 days


1/A + 1/B = 2/15 or (A+B)/AB= 2/15 ----(i)


(1/2)/(1/A) + (1/2)/(1/B) = 20 or A/2 + B/2 = 20 or A+B= 40 ----(ii)


Substituting A+B= 40 from (ii) in (i), we get AB= 300


Now finding the factors of 300, whose sum is 40, we have 10 and 30


As B is more efficient than A, B can do the work in 10 days.
```

Question 5 (PrepInsta quiz-1 - Q5)

Question Stem:
An inlet pipe fill a tank in 5 hrs and outlet pipe empty same tank in 36 hrs working individually.how many additional number of outlet pipes of same capacity required to be opened so that tank never overflows?

Options:
- **(A)** `4`
- **(B)** `8`
- **(C)** `7`
- **(D)** `10`

Correct Answer: (C) `7`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Inlet pipe fills the tank in 5 hr and outlet pipe empties the tank

in 36 hr.

So inlet pipe is 36/5 = 7.2 times efficient than , ensure that the

an outlet pipe.

Therefore , in order to tank never overflows , we will need total

8 outlet pipes.

Already we have 1 outlet pipe , thus we need only (8-1) = 7 outlet pipes
```

Question 6 (PrepInsta quiz-1 - Q6)

Question Stem:
Shanti's school normally FINISHES AT 4 PM. her mom drives from home to pick her up, reaching the school exactly at 4 pm. one day, a half-holiday is announced and the School finishes for the day at 1 pm. Rather than sitting and Waiting, Shanti decides to start walking towards home. Her mother meets her along the way and as a result they reach home an hour earlier than normal. what is the ratio of the Shanti's walking speed to her mother's driving Speed?

Options:
- **(A)** `1:5`
- **(B)** `3:9`
- **(C)** `7:10`
- **(D)** `3:5`

Correct Answer: (A) `1:5`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Consider this way....

Shanti walked and as a result she saved the total driving hours= 1

Which means her mom saves half an hour each going up and

down since Santi met her on the way.

That means both

met at 3.30 pm. So Shanti walked what the distance in 2.5 hrs

(from 1 pm to 3.30 pm) what her mom would have driven for half an hour.

So the ratio of their speeds are 0.5: 2.5 i.e. 1:5.

The clue lies in considering the to and fro distance.

Hope it is clear now.

:)
```

Question 7 (PrepInsta quiz-1 - Q7)

Question Stem:
In a grass field if 40 cows could eat for 40 days.The same grass field can feed 30 cows for 60 days.how long will it feed 20 cows?

Options:
- **(A)** `80`
- **(B)** `85`
- **(C)** `70`
- **(D)** `60`

Correct Answer: (A) `80`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
80

Here if number of cows decrease by 10 has increasing the

number of days by 20

therefore,

40cows for 40days

30cows for 60days

20cows for 80days

Ans: 80

or you can solve by ratio

40C : 40D --&gt; 1:1

30C : 60D --&gt; 1:2

20C : 80D --&gt; 1:4

ans:80
```

Question 8 (PrepInsta quiz-1 - Q8)

Question Stem:

5 skilled workers can build a wall in 20days; 8 semi-skilled workers can build a wall in 25 days; 10 unskilled workers can build a wall in 30days. If a team has 2 skilled, 6 semi-skilled and 5 unskilled workers, how long will it take to build the wall?

Options:

- **(A)** `12`
- **(B)** `15`
- **(C)** `14`
- **(D)** `18`

Correct Answer: (B) `15`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Ans : 15days


For Skilled


5 workers --------20 days


5 workers 1 day work= 1/20


1 worker's 1 day work= 1/(5*20)


Similarly,


For Semi-Skilled------


1 worker's 1 day work= 1/(8*25)


For Unskilled------


1 worker's 1 day work= 1/(10*30)


For 2 skilled,6 semi-skilled and 5 unskilled


workers


One day work= 2*[1/(5*20)] +6*[1/(8*25)] + 5*[ 1/(10*30)] = 1/15


Therefore no. of days taken= 15
```

Question 9 (PrepInsta quiz-1 - Q9)

Question Stem:

Rajesh can finish 1/5 of his home work in one hour. Seema can finish 3/7 of her homework in 90 minutes and Ramya can finish 3/4 of her homework in three and a half hours. If all of them start their home work at 12.00 PM and can go to play as soon as they all finish their homework. When can they start to play, if they take a break at 3.30 PM for 30 minutes?

Options:

- **(A)** `5:10 pm`

- **(B)** `6:30 pm`
- **(C)** `5:30 pm`
- **(D)** `5:45 pm`

Correct Answer: (C) `5:30 pm`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Starting homework at 12 PM and Taking a break at 3:30 PM,

we came to know that each of them have already spent

3 hrs and 30 minute i.e 210 min

After taking break, they started to do their homework at 4 PM

1. Rajesh can finish 1/5 of his work in 1 hr. i.e in 60 min

He can finish 1 work in 60* 5 = 300 min

Remaining time to complete his hm work = 300-210= 90 min

Starting again at 4 PM,he will finish his homework at 5:30 PM

2. Sema can finish 3/7 of her homework in 90 min

She will finish 1 work in 90*(7/3) = 210 min

Therefore she finished her work at 3:30 PM

3.

Ramya can finish 3/4 of her work in 210 min

She will finish 1 work in 210*(4/3 ) min= 280 min

Remaining time left for Ramya to complete her work =280 - 210 = 70 min

Starting to do homework again at 4 PM,she will complete her work at

5:10 PM

Since Rajesh takes longer time to finish the work

(completing the work at 5:30 PM), hence they all can start
```

```
to play at 5:30 PM. That  the only time they all will meet
```

Question 10 (PrepInsta quiz-1 - Q10)

Question Stem:
3 person can fill tank in 25 min, a can fill in 30 min , b can fill in 35 min and c can empty the 5 gallon per min then what is the capacity of tank?

Options:
- **(A)** `230`
- **(B)** `250`
- **(C)** `200`
- **(D)** `180`

Correct Answer: (A) `230`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
the answer is approximately 230 gallons and if u want exact then it comes 228.25


explanation is given below:


3 person together fill tank in 25 min hence, calculate the time of person C that in how much time
he empty the tank so,


(1/A)+(1/B)-(1/C)=1/25


i.e, (1/30)+(1/35)-(1/C)=1/25 .......... we get,


1/C=23/1050,


while, its given the C can empty 5 gallon per min. therefore


capacity of tank = 5*1050/23


and capacity of tank will get if we solve above expression,


we get 228.25
```
